Countries of the world agree that something needs to be done about these important environmental issues. Many global summits, conferences and conventions are regularly conducted by the United Nations and many steps are taken to minimise human-induced issues by signing agreements with around 150 countries.

Activity

Students may form 'ECOGROUPS' and discuss eco-issues of their premises and find solutions to the existing problems like, litter disposal, water stagnation, health and hygiene, greening the campus and its maintenance.

Drastic increase in population resulted in demand for more productivity of food materials, fibres, fuels which led to many environmental issues in agriculture, land use modifications resulting in loss of biodiversity, land degradation, reduction in fresh water availability and also resulting in man-made global warming by green house gases even altering climatic conditions.

8.1 Green House effect and Global Warming

Green House Effect is a process by which radiant heat from the sun is captured by gases in the atmosphere that increase the temperature of the earth ultimately. The gases that capture heat are called **Green House Gases** which include carbon dioxide (CO_2), methane (CH_4), Nitrous Oxide (N_2O) and a variety of manufactured chemicals like chlorofluorocarbon (CFC). Increase in greenhouse gases lead to irreversible changes in major ecosystems and climate patterns. For example, coral ecosystem is affected by increase in temperature, especially **coral bleaching** observed in Gulf of Mannar, Tamil Nadu.

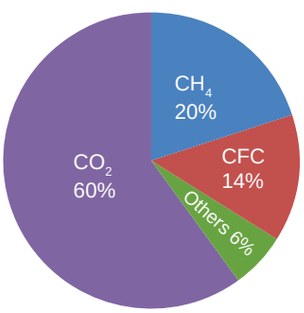


Figure 8.2: Relative contribution of green house gases

Human activities lead to produce the green house effect by

- Burning fossil fuels, which releases CO_2 and CH_4
- Way of Agriculture and animal husbandry practices
- Electrical gadgets like refrigerator and air conditioners release chloro fluoro carbons
- The fertilizers used in Agriculture which release N_2O
- The emissions from automobiles.

The increase in mean global temperature (highest in 4000 years) due to increased concentration of green house gases is called **global warming.**

One of the reasons for this is over population which creates growing need for food, fibre and fuel and considered to be the major cause of global warming.



Clouds and Dust particles can also produce Green House effect. That is why clouds, dusts and humid nights are warmer than clear dust free dry nights.

8.1.1. Effects of Global Warming

- Rise in global temperature which causes sea levels to rise as polar ice caps and glaciers begin to melt causing submergence of many coastal cities in many parts of the world.
- There will be a drastic change in weather patterns bringing more floods or droughts in some areas.
- Biological diversity may get modified, some species ranges get redefined. Tropics and sub-tropics may face the problem of decreased food production.

TN_GOVT_BIOLOGY-BOTANY_XII_PAGES_169-184 CH 08.indd 156 3/14/2020 11:48:18 AM